Curriculum Vitae

PERSONAL INFORMATION



First name: Muhammad Asad, **Family name:** Lal
Email: asad@njust.edu.cn
WeChat ID: iamAsadLal
Research Profile: asadlal.tech
Date of birth: 26 January 1998
Nationality: Pakistani

AREAS OF RESEARCH

Metaheuristic and Evolutionary Optimization; Artificial Intelligence; Medical Image Analysis; Engineering Design Optimization; Machine Learning; Deep Learning

EDUCATION

08/2024 – 04/2027 Master (Computer Science and Technology)
School of Computer Science and Engineering,
Nanjing University of Science and Technology, China

08/2012 – 01/2015 Bachelor (Computer Science)
Department of Computer Science,
Government College University Faisalabad, Pakistan

PUBLICATIONS *Google Scholar*

1. **Lal, M.A.**, Kashif, M. & Huang, B. Father-inspired metaheuristic: a hybrid framework with neighborhood repulsion and elite learning for continuous optimization. ***Evol. Intel.* 19, 111 (2026).**
2. **Lal, M.A.**, Kashif, M., Lv, J. & Huang, B. AFRC-SHADE: Augmented Forman-Ricci Curvature-Guided SHADE for Multi-Level Thresholding in Fundus Image Segmentation. ***Swarm and Evolutionary Computation. Submitted* (2026).**

RESEARCH WORK

08/2024 – 04/2027 Development of Metaheuristic Algorithms
- Developed metaheuristic algorithms for continuous optimization.
- Evaluated performance using benchmark functions and statistical analysis.

08/2024 – 04/2027 Medical Image Analysis
- Applied metaheuristic optimization to medical image segmentation.
- Evaluated segmentation using quality metrics: PSNR, SSIM, and FSIM.

PROFESSIONAL EXPERIENCE

10/2023 – 03/2024 Android Development, Internship
Idealogic Software House, Punjab, Pakistan

UNDERGRADUATE PROJECTS

08/2019 – 08/2023 Heart Disease Prediction, Final Year Project
- Developed an Android app for heart disease prediction.
- Used Logistic Regression for health classification.

02/2023 – 05/2023 Deep Learning-based Face Recognition, Semester Project
- Developed a CNN-based face recognition model.
- Performed face recognition using image data.

FELLOWSHIPS AND AWARDS

08/2024 Fully Funded High Achievers (Lanshan) Scholarship
Nanjing University of Science and Technology, China

08/2023 Outstanding Academic Performance Award
Government College University Faisalabad, Pakistan

CERTIFICATIONS

06/2026 One Million Prompters
Dubai Future Foundation

12/2024 Chinese Writing Competition
School of International Education, Nanjing University of Science and Technology

04/2024 Manage Sensitive Data and Security Policies within GitHub
Microsoft

11/2023 Machine Learning with Python
International Business Machines (IBM)

11/2023 Deep Learning with TensorFlow
International Business Machines (IBM)

10/2017 Computer Application and Office Professional
Government of Punjab Vocational Training Council, Pakistan

TECHNICAL SKILLS

- Python
- C/C++
- LaTex
- MATLAB
- VS Code
- GitHub

LANGUAGE

- English – Fluent
- Chinese – Basic
- Urdu – Native
- Arabic – Basic

EXTRA-CURRICULAR ACTIVITIES

03/2026 Xuanwu Youth Development Activity
Xuanwu District Government, Nanjing, China

08/2025 International Student Registration Volunteer
Nanjing University of Science and Technology, China

06/2023 Basic Community Safety Training
Punjab Emergency Service Rescue 1122, Pakistan

05/2022 BSAFE Security Awareness Training
United Nations Department of Safety and Security

REFERENCE

Prof. Bo Huang
School of Computer Science and Engineering,
Nanjing University of Science and Technology, China
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